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Procedia Computer Science 187 (2021) 241–245

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Identification, Information and Knowledge in the internet of Things, 2020

Study on Absorbing Characteristics of Salisbury Screen Filled with Plasma

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Abstract

Salisbury screen is the basic type of absorbing structure, which has the defects of fixed absorbing characteristics and narrow absorbing frequency band. This paper proposes a plasma-filled Salisbury screen and simulates its wave-absorbing characteristics. The simulation results show that the plasma-filled Salisbury screen can adjust the absorbing center frequency of the Salisbury screen by adjusting the plasma resonance frequency and electron collision frequency, and its absorbing frequency bandwidth is higher than that of the traditional Salisbury screen.

Keywords: Salisbury screen; Plasma; Wave-absorbing characteristics;

1. Overview

The confrontation between radar and stealth technology has become one of the most concerned military technologies today. From the perspective of stealth, in order to reduce the RCS of the target, the following technical measures are often combined: (1) Shape technology, that is, to reduce RCS from the shape design; (2) Absorbing material technology, that is, in the strong scattering area or location Coating absorbing materials to reduce RCS; (3) Impedance loading technology, that is, on the appropriate part of the aircraft, without affecting the aerodynamic performance and structural strength of the aircraft, increase the distributed active or passive loading impedance to Add additional scattered field to offset or reduce the scattered field of the aircraft, thereby achieving the purpose of reducing RCS[1][2].

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10.1016/j.procs.2021.04.057

At present, radar absorbing material (RAM) technology is one of the fastest growing and most widely used technologies in electromagnetic stealth. The development and application of RAM is the key to the development of stealth technology. According to the material forming process and carrying capacity, radar absorbing materials can be divided into coated absorbing materials and absorbing structures. Coating type absorbing material is to form a absorbing coating by mixing the binder with metal, alloy powder, ferrite, conductive fiber and other absorbents. Absorbing structure is a dual-functional composite material developed on the basis of advanced composite materials. It can absorb waves and carry loads. It has the advantages of wide frequency bandwidth, high efficiency, and no increase in negative weight. It is the main direction of research and application of absorbing materials. The Salisbury screen[3], Jaumann[4] and Dallenbach absorbers that appeared after the 1950s are all early absorbing structures.

The Salisbury screen consists of a resistive layer, a filling layer and a conductive layer. The resistive layer is placed in front of the conductive plate and separated by a spacer layer with low loss/lossless, low dielectric constant and relative admittance. The resistance layer and the spacer layer, and the spacer layer and the conductor layer are generally connected by pasting. The thickness of the resistive layer is much smaller than the wavelength, generally from micrometers to sub-millimeters; the thickness of the spacer layer is an odd multiple of the wavelength of the 1/4 incident wave, generally 1/4 of the incident wavelength; the main function of the conductor layer is to reflect electromagnetic waves, There is no specific requirement for thickness. The disadvantage of the traditional Salisbury screen is that once it is manufactured, its characteristics are fixed. Therefore, if the threat is changed, the absorbing performance under the new threat will decrease. In addition, from the absorbing mechanism of the Salisbury screen, it can be seen that the electrical scale of the interface, the dielectric constant of the material, and the permeability are all functions of frequency; according to the Rozanov limit[5], under a certain thickness constraint, the work of the Salisbury screen Bandwidth is always limited. In order to solve this problem, this paper proposes a plasma-filled Salisbury screen based on the Salisbury screen. The absorbing bandwidth and absorbing center frequency of the absorbing structure are adjusted by adjusting the resonant frequency and electron collision frequency of the filled plasma.

2. Salisbury screen and its absorption characteristics

The Salisbury screen is a resonant absorber. Its structure is to place a resistor piece at a distance in front of the metal plate. The unit of surface resistance of the resistor piece is Ω*cm^2, and the unit of equivalent resistance in its equivalent circuit is represented by Ω.

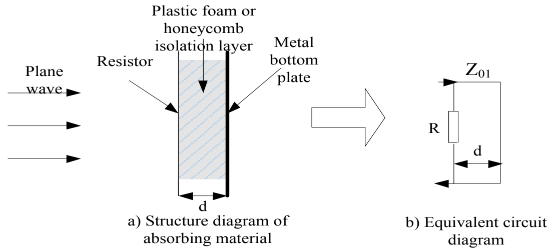

Fig 1 Salisbury screen absorber and its equivalent circuit

In Figure 1a, the relative dielectric constant of the plastic foam or honeycomb isolation layer is between 1.03 and 1.1, and the relative permeability is 1. For the convenience of analysis, the relative dielectric constant is regarded as 1, and the equivalent resistance is R, the conductance is G=1/R, and the resistor piece is regarded as infinitely thin. The transmission line theory is used to analyze the reflection characteristics of the Salisbury screen[6], and the equivalent circuit of the absorbing material shown in Figure 1a is shown in Figure 1b.

First, find the input impedance of a transmission line with a length of d and a characteristic impedance of Z_{01}:

$$Z_{in} = Z_{01} \frac{Z_l + jZ_{01} \tan \beta d}{Z_{01} + jZ_l \tan \beta d} \tag{1}$$

Where β is the free space wave number, Z_l is the load impedance of the transmission line, where Z_l=0, so the above formula is converted to:

$$Z_{in} = jZ_{01} \tan \beta d \tag{2}$$

After connecting in parallel with the resistor R, the input impedance seen from the beginning is:

$$Z'_{in} = \frac{Z_{in} R}{Z_{in} + R} \tag{3}$$

Then the reflection coefficient is:

$$\Gamma = \frac{Z'_{in} - Z_0}{Z'_{in} + Z_0} \tag{4}$$

To make the reflection coefficient zero, there must be $Z'_{in} = Z_0$, which can be obtained by formula (3):

$$\frac{1}{Z_0} = \frac{1}{R} + \frac{1}{Z_{in}} \tag{5}$$

After converting to admittance:

$$Y_0 = G + Y_{in} \tag{6}$$

Where: G——Parallel admittance;

Z_0——Characteristic impedance of free space;

Y_0——Characteristic admittance of free space, Y_0=1/ Z_0;

Y_{in}——The input admittance of a transmission line of length d.

Since $Y_{in} = \dfrac{1}{jZ_{01} \tan \beta d}$, G and Y_0 are real numbers, it can be seen from equation (6) that when $G = Y_0$, the condition for Z'_{in} to match the characteristic impedance of free space is $Y_{in} = 0$, then:

$$\beta d = \frac{\pi}{2} + n\pi \quad n = 0,1,2\cdots \tag{7}$$

And $\beta = \dfrac{2\pi}{\lambda_0}$, λ_0 are free-space wavelengths, and d should be:

$$d = \frac{\lambda_0}{4} + \frac{n\lambda_0}{2} \quad n = 0,1,2\cdots \tag{8}$$

It can be seen from formula (8) that the thickness of the isolation layer of the Salisbury screen must be an odd multiple of a quarter wavelength in order to match the input impedance of the entire structure with the free-space characteristic impedance, thereby achieving zero reflection.

A calculation example is given below. Take R=377Ω/, the center frequency is 10GHz($\lambda_0 = 30mm$), and the thickness of the isolation layer is 7.5mm. The calculation result is shown in Figure 2.

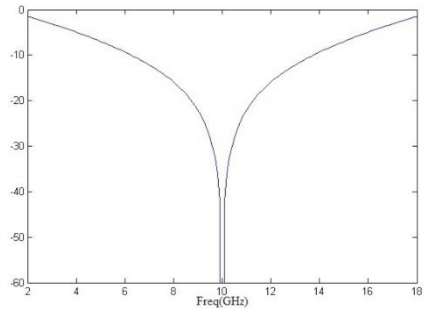

Fig 2 Reflective characteristics of Salisbury screen with isolation layer thickness of 7.5mm

3. Salisbury screen filled with plasma

When analysing the propagation of electromagnetic waves in plasma, the plasma is usually regarded as a medium equivalent. The carrying current generated by the electrons and ions in the plasma under the action of the electric field will determine the equivalent dielectric constant of the plasma. When there is no external magnetic field, and considering the collision of particles in the plasma, the relative dielectric constant of the non-magnetized plasma is

$$\varepsilon_r = 1 - \frac{\omega_p^2}{\omega(\omega - jv)} = \left(\frac{(j\omega)^2 + v(j\omega) + \omega_p^2}{(j\omega)^2 + v(j\omega)} \right) \tag{9}$$

The relative dielectric constant is a variable that changes with the frequency of the incident wave. The electromagnetic transmission characteristics of the plasma are affected by the combination of three factors: the frequency of the incident wave, the resonance frequency of the plasma, and the frequency of electron collision. Therefore, in the Salisbury screen, the spacer layer between the resistor plate and the metal bottom plate is replaced with a plasma layer, and the wave absorbing ability and the wave absorbing centre frequency of the Salisbury screen can be adjusted by adjusting the resonance frequency of the plasma and the electron collision frequency.

The equivalent circuit method is used to simulate and analyse the plasma-filled Salisbury screen. In the simulation, the plasma layer thickness d=7.5mm, the equivalent resistance of the impedance layer is 377Ω, and the electron collision frequency v=0, respectively calculate the reflectivity of the absorber when the plasma resonance frequency ωp1=6GHz, ωp2=10GHz and ωp3=15GHz. For the convenience of comparison, the figure also shows the reflectivity of the wave absorbing device filled with foam material with electromagnetic properties similar to free space. It can be seen from the figure that the central absorbing frequency of the wave absorbing device can be adjusted by adjusting the plasma resonance frequency, and the absorbing bandwidth of the wave absorbing device filled with plasma is significantly wider than that of the wave absorbing device filled with foam.

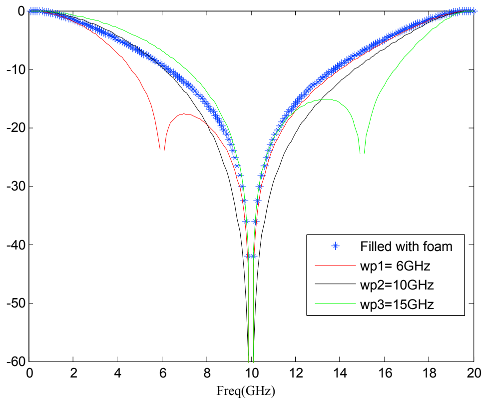

Fig 3 Salisbury screen reflection coefficient filled with plasma

4. Conclusion

According to the results of simulation calculations, the plasma-filled Salisbury screen has the effect of broadening the absorption frequency band, and adjusting the centre frequency and width of the absorption frequency band by adjusting the collision frequency and resonance frequency of the plasma. Since the current Salisbury screen and its derivative wave absorbing structure have been widely used, the plasma-filled Salisbury screen has a wider application prospect.

The density distribution of the filling layer plasma currently considered is uniform. In fact, the filling layer plasma density distribution is non-uniform, which is related to many parameters of the plasma generator. Therefore, simulation and experimental verification of the absorbing characteristics of the Salisbury screen under the real plasma density distribution are the main work directions in the next step.

Acknowledgements

This work is supported by the National Natural Science Foundation of China under Grant (No.11927803).

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